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Personal big data pricing method based on differential privacy☆

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ABSTRACT

Personal big data can greatly promote social management, business applications, and personal services, and bring certain economic benefits to users. The difficulty with personal big data security and privacy protection lies in realizing the maximization of the value of personal big data and in striking a balance between data privacy protection and sharing on the premise of satisfying personal big data security and privacy protection. Thus, in this paper, we propose a personal big data pricing method based on differential privacy (PMDP). We design two different mechanisms of positive and reverse pricing to reasonbly price personal big data. We perform aggregate statistics on an open dataset and extensively evaluated its performance. The experimental results show that PMDP can provide reasonable pricing for personal big data and fair compensation to data owners, ensuring an arbitrage-free condition and finding a balance between privacy protection and data utility.

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1. Introduction

With the advent of the era of artificial intelligence, personal big data have great value in the data market. The term "personal big data" refers to private data generated in the daily life or work of individuals, who have ownership of those data (Shen et al., 2016). With the further development of the mobile Internet, personal big data as a valuable asset are growing exponentially (Guo et al., 2017). However, large amounts of potentially useful private data cannot be accessed by stakeholders because of privacy concerns (Haddadi et al., 2012). In particular, many organizations benefit from collecting, selling, and analyzing personal big data. Moving away from the narrow view of data confidentiality and monetizing personal big data will allow individuals to control their data in monetary

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terms. The privacy attribute of personal big data introduces a number of challenges, and different ways of dealing with this commodity is required. How much money should data buyers pay? and how much money should data owners charge for their privacy loss resulting from the use of information derived from their personal big data? One possible way is to assign a price commensurate to the amount of privacy loss; however, how to quantify privacy loss and how much money to pay for a metric of privacy loss are the radical challenges in this market.

To develop a personal big data pricing method based on differential privacy (PMDP), we need to address three major challenges. The first challenge is how to quantify the loss of privacy. To mine the value of personal big data and improve its practicability, we find that individual users will inevitably suffer from different degrees of privacy loss. How to quantify the loss of privacy suffered by the individuals and how to compensate them for it are key technical problems. At the same time, there is a correlation between data, and this correlation should be fully considered in the process of aggregate statistics. How to provide fair and reasonable compensation to individual users according to the degree of correlation between the data and how to allocate compensation among different users will become important.

The second challenge is how to trade personal big data fairly while protecting privacy. In other words, how should a reasonable trading framework be designed to prevent circumvention from buyers on arbitrage pricing and from data owners on untruthful privacy valuation? Just as Facebook (2018) and Twitter (2018), once involved the disclosure of personal privacy, it is necessary for individuals to protect their personal big data before participating in the transaction, and differential privacy protection can be achieved by adding noise interference. Adding different degrees of noise variance (Cummings et al., 2015) can achieve different degrees of privacy protection, obtain data with different levels of accuracy, and make the data have a different utility and price. The higher the noise, the lower the accuracy, the lower the price, and vice versa. More importantly, adding noise makes it more difficult for attackers to arbitrage their pricing of data. Free arbitrage is important for a robust market of personal data.

The third challenge is how to establish a balanced pricing mechanism. That is, how much are personal big data worth? How does one strike a balance between the loss of privacy for the data owner and the payment to the buyer? This balance is crucial in convincing data owners and data buyers to participate in the personal big data market.

Here is a summary of our contributions.

- We propose PMDP, and PMDP stands for pricing method based on differential privacy.
- We propose a balanced pricing mechanism that computes the price of a noisy aggregated query answer and compensates the data owner according to the degree of privacy loss. Our mechanism will empower both data owners and data buyers to fully control their own benefits and expenses.
- We design two different mechanisms of positive and reverse pricing to reasonably pricing personal big data.
- PMDP can make different pricing according to different variances.
- PMDP can compensate individuals for their privacy losses using both positive pricing and reverse pricing.
- We perform aggregate statistics on a real dataset and evaluate their performance extensively.

2. Pricing model and key principles

This section describes the personal big data pricing model based on differential privacy and its key technical principles.

2.1. *Pricing model*

This section first defines the personal big data pricing model based on differential privacy. Fig. 1 shows an illustration of the model. The model consists of three roles: data owner, data buyer, and personal data bank.

Personal big data Personal big data refers to behavior data with individual characteristics generated in personal life activities or work, which can be owned or controlled by individuals. The sources of personal big data are complex and diverse, including personal basic information, personal income, personal property, personal circle of friends, personal health, personal education, personal diaries, personal documents, personal opinions, personal perception information, etc. Shen et al. (2016).

Personal data bank The personal data bank can carry out asset management and value-added services for personal big data, and its core principle is to protect the ownership, right to know, privacy, and usufruct of personal big data (Guo et al., 2017). The personal data bank organizes all data centered on individuals, and it provides some key functions of ownership verification, aggregation, management, trading, and value-added services for personal big data, which can effectively guarantee that all data access is authorized and that all data circulate orderly. Data pricing is the key technology in data trading. The personal data bank collects data from individual users, adds the appropriate noise according to the privacy requirements of individual users, protects data privacy, analyzes and mines data, and sells data to buyers in the form of a query. The purchaser shall issue a data inquiry request to the personal data bank and pay the relevant fees. The personal data bank compensates individual users for their loss of privacy.

Data owner The data owner, namely, the individual user, provides tuple data to the personal data bank. Tuples consist of different attributes of personal data. The dot product of a property field is the possible set of all tuples that appear. The personal data bank collects the personal tuple data to form a relational table.

Definition 1 (Dataset)**.** A vector consisting of n numeric data items represents the dataset, i.e., $D = (d_1, d_2, \ldots, d_n)$.

A data item d_i can indicate the personal rating of a movie on a scale of 1 (worst) to 5 (best). Data items can indicate a patient's hepatitis status, such as $d_i = 0$ for negative or $d_i = 1$ for positive.

Buyer The data buyer needs the personal data bank to conduct a linear aggregate query of some personal data, obtain

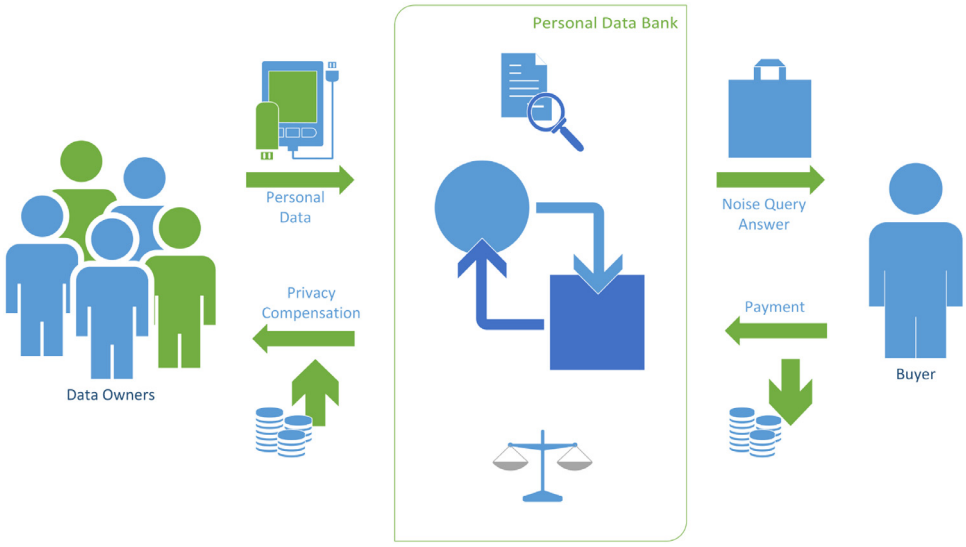


Fig. 1 – Illustration of the personal big data pricing model based on differential privacy.

the required query results, and pay the purchase fee to the personal data bank.

Definition 2 (Linear Query Chao et al. (2014)). Let the vector $q = (q_1, q_2, \ldots, q_n)$ represent a linear query. The dot product $qD = q_1d_1 + q_2d_2 + \ldots + q_nd_n$ represents the result of a linear query $q(D)$ on the dataset D.

Common linear queries include the basic aggregate statistics such as the sum, average, and count, as well as grouping queries and data cube graph queries. Linear queries can be used for statistical histograms, marginal costs, or complex tasks such as decision trees and log-linear models.

Ganzhao et al. (2015) proposed an optimization framework that minimizes the overall error of the results for a batch of linear queries under defferential privacy. Xiaoguang et al. (2019) focused on the linear query function in Laplace based mechanisms, and prposed a method to set the upper bound for the number of linear queries from the perspective of information theory. Parameswaran et al. (2020) developed new d_x-private mechanisms for linear queries by extending the standard ε-differentially private mechanisms.

To protect personal privacy, we adopt a differential privacy method to protect personal privacy information. To prevent buyers from arbitraging, namely, from combining known queries to derive other query results, we require any direct or indirect private information to be paid for according to the correlation between the data.

2.2. *Key principles*

To design a reasonable trading framework and a balanced pricing mechanism, we must determine the main principles of personal big data pricing. The principles are categorized into four different groups: differential privacy, dependent differential privacy, query type and answerability, and arbitrage free. To guarantee the data owner's privacy, differential privacy injects some randomness into the result on the basis of the preferred privacy level. It is also used as a metric to quantify the privacy loss of each data owner. With this differential privacy guarantee, only certain linear aggregated query types are applicable in this trading framework. Regarding pricing, a pricing method should be arbitrage free and must not allow any circumventions on the query price from any savvy buyers. Similarly, such a framework should be designed to encourage data owners' truthful privacy valuation by providing them with the right pricing scheme so that they will not benefit from any untruthful valuation. Finally, it is important to ensure the generation of an unbiased/less biased query result without increasing the query price. Therefore, a careful sample selection method is crucial.

2.2.1. *Differential privacy*

The pricing mechanism should be capable of preserving the data owner's privacy from any undesirable privacy leakages. To ensure privacy, differential privacy (Dwork and Roth, 2014) plays an essential role in guaranteeing that the adversary does not know any individual private information but just the whole dataset from observing the query result (despite some background knowledge about that individual). Given a privacy parameter ε, any private mechanisms (e.g., Laplace mechanism and exponential mechanism) satisfy the ε-differential privacy level if the same result is likely to occur whether a item exists in the dataset or not as a result of random noise addition. A smaller ε offers better privacy protection but is less accurate, resulting in a tradeoff between privacy and result accuracy. In our pricing method, we define ε as the quantification of the privacy loss of the data owner because ε and money are correlated.

If two datasets have only one item that is different, the two datasets are considered adjacent. The concept of adjacent datasets is useful for understanding differential privacy. Here is the formal definition.

Definition 3 (Neighboring datasets). Let D, D' denote two datasets; if $D \subset D'$ and $|D'| = |D| + 1$, then we can say that they are neighboring each other, denoted as $D \sim D'$.

$D \overset{t}{\sim} D'$ indicates that there is only one item that is different between the two datasets, i.e., $t \in D', t \notin D$.

Differential privacy is a method in cryptography designed to provide a way to maximize the accuracy of data queries when queried from a statistical database, while minimizing the opportunity to identify its records. With the notion of neighboring datasets in mind, a formal definition of ε-differential privacy can be given below.

Definition 4 (ε-Differential Privacy Dwork and Roth (2014)). A random algorithm M satisfies ε-differential privacy (ε-DP) if the neighboring datasets D and D' differ by only one item and any possible output of $O \subseteq Range(M)$, which can be expressed as formula Eqn 1:

$$Pr(M(D) \in O) \leq exp(\varepsilon) \cdot Pr(M(D') \in O) \tag{1}$$

Differential privacy can be achieved by adding random noise to the output of the function $f : D \rightarrow R$, and the noise size can be adjusted according to the global sensitivity or to the maximum impact of the input items on the output.

Definition 5 (Global Sensitivity Dwork et al. (2017)). Let $f : D \rightarrow \mathbb{R}^d$ represent a function, whose global sensitivity can be expressed as formula Eqn 2:

$$\Delta f = \max_{D \sim D'} \| f(D) - f(D') \|_1 \tag{2}$$

where $\| \cdot \|_1$ is a L_1 norm.

Let $Lap(\frac{\Delta f}{\varepsilon})$ represent the Laplace distribution with mean 0 and scale $\frac{\Delta f}{\varepsilon}$; then, we can add random noise to it to construct a function f satisfying the ε-differential privacy.

Theorem 1 (Laplace Mechanism Dwork et al. (2017)). *For a function $f : D \rightarrow \mathbb{R}^d$, the mechanism that returns $f(D) + z^d$, where each z_i is drawn independently and identically distributed from $Lap(\frac{\Delta f}{\varepsilon})$, satisfies the ε-differential privacy,*which can be expressed as formula Eqn 3:

$$M(D) = f(D) + Lap(\frac{\Delta f}{\varepsilon}) \tag{3}$$

Another implementation mechanism for differential privacy is the exponential mechanism (Mcsherry and Talwar, 2007) for functions where the output space is non-numeric or adding noise makes no sense.

Definition 6 (Score Function Jorgensen et al. (2015)). Assuming that D and D' have only one item that is different, denoted as $D \oplus D'$, then we can express the score function as formula (4):

$$S(D, r) = \max_{f(D')=r} -|D \oplus D'| \tag{4}$$

The higher the score of the score function is, the better the value of r is relative to $f(D)$, i.e., the closer it is to the real result.

Theorem 2 (Exponential Mechanism Mcsherry and Talwar (2007)). *For a score function $s : D \times R \rightarrow \mathbb{R}$, the mechanism $\varepsilon_\varepsilon^s(D)$ that outputs $r \in R$ with a probability proportional to $exp(\frac{\varepsilon s(D,r)}{2\Delta_s})$ satisfies the ε-differential privacy.*

The global sensitivity of the score function can be expressed as $\Delta_s = \max_{D \sim D', r \in R} |s(D, r) - s(D', r)|$.

2.2.2. *Dependent differential privacy*

The dependent differential privacy principle (Liu et al., 2016) comes from the well-known differential privacy principle (Dwork and Roth, 2014), which can also be used to quantify the privacy loss of the data owner. The difference is that the former cares about the correlation between the data and can more accurately quantify the privacy loss of the individual. The specific technical notations are shown below.

For any dataset $D = (d_1, d_2, \ldots, d_n)$, we define its dependence size to be L if any item in D is dependent on at most $L-1$ other items. We denote by R the probabilistic dependence relationship among the L dependent items. Relationship R could be result from the data generating process or other social, behavioral and genetic relationships arising in real-world scenarios. The dependent neighboring datasets is defined as follows.

Definition 7 (Dependent Neighboring Datasets). Let $D(L, R)$ and $D'(L, R)$ denote two dependent datasets; if $D(L, R) \subset D'(L, R)$ and $|D'(L, R)| = |D(L, R)|+1$, then we can say that they are neighboring each other, denoted $D(L, R) \sim D'(L, R)$.

$D(L, R) \overset{t}{\sim} D'(L, R)$ indicates that there is only one item that is different between the two dependent datasets, i.e., $t \in D'(L, R)$, $t \notin D(L, R)$.

Depending on the dependent neighboring datasets, the formal definition of dependent differential privacy is as follows.

Definition 8 (ε-Dependent Differential Privacy (Liu et al., 2016)). Let $D(L, R)$ and $D'(L, R)$ denote two dependent neighboring datasets, and let O denote any possible output; then, a mechanism M that provides ε-dependent differential privacy has to satisfy formula Eqn 5:

$$Pr(M(D(L, R)) \in O) \leq exp(\varepsilon) \cdot Pr(M(D'(L, R)) \in O) \tag{5}$$

where ε represents the privacy budget. The smaller the value of ε, the better the privacy protection, the worse the utility, and the lower the data price.

Liu et al. (2016) further quantified the dependencies between data items in a fine-grained manner by adding a dependency interference mechanism such as Laplace noise between the data items to realize the dependent differential privacy. ρ_{ij} is used to represent this fine-grained dependency between d_i and d_j, which quantifies the change caused by modifying d_i to d_j and is expressed by formula (6):

$$\rho_{ij} = \frac{\max_{d_i} log\{\sum_{d_j} P(D_j = d_j | D_i = d_i) exp(\frac{\|d_j - d_j^*\|_1}{\sigma(\varepsilon)})\} \sigma(\varepsilon)}{\Delta D_j} \tag{6}$$

where d_j^* represents the optimal solution of $\underset{d_{j1}}{argmax} \|d_j - d_{j1}\|_1$, ΔD_j denotes the maximal difference in D_j, and $\sigma(\varepsilon) = \frac{\Delta D_j}{\varepsilon}$. $P(D_j = d_j | D_i = d_i)$ denotes the probability of a user j being located at d_j when his/her friend i is located at d_i, which can be expressed as formula (7):

$$P(D_j = d_j | D_i = d_i) = a(\|d_j - d_i\|_1 + b)^{-c} \tag{7}$$

Formula (7) was used to calculate the dependence coefficient of formula (4), and $a = 0.0019$, $b = 0.196$, and $c = 1.05$ were set according to reference (Backstrom et al., 2010).

The dependency coefficient ρ_{ij} evaluates the degree of dependence between data items from the perspective of privacy, and its value range is greater than or equal to 0 and less than or equal to 1.

Let f represent a query function; then, the dependence sensitivity of f on d_j caused by the modification of d_i can be calculated by formula (8):

$$DS_{ij}^{f} = \rho_{ij}\Delta f_j \tag{8}$$

Here, $\Delta f_j = \max_{d_{j1},d_{j2}} \parallel f(\ldots, d_{j1}, \ldots) - f(\ldots, d_{j2}, \ldots) \parallel_1$ indicates the sensitivity caused by modifying d_j.

When there are multiple users, dependent sensitivity can be expressed by formula (9):

$$DS_{i}^{f} = \sum_{j\in\mathbb{C}_i} \rho_{ij}\Delta f_j \tag{9}$$

Here, $\mathbb{C}_i$ represents a collection of L items dependent on the ith item and $\rho_{ii} = 1$. DS_i^f is used to measure the sensitivity of f to all items in the dataset D due to the modification of the ith item.

Theorem 3. *The dependency sensitivity of any query function f published on the dependent dataset is shown in formula (10):*

$$DS^{f} = \max_{i} DS_{i}^{f} \tag{10}$$

The dependency perturbation mechanism (DPM) is formally defined below by formula (11) for the implementation of the ε-dependency differential privacy.

Theorem 4. *For any query function f, mechanism M on dependent dataset D can be expressed by formula (11):*

$$M(D) = f(D) + Lap(\frac{DS^{f}}{\varepsilon}) \tag{11}$$

which gives the ε-dependent differential privacy.

2.2.3. Query type and answerability

With some background knowledge, the adversary may engage in linkage attacks on the published query answer and may eventually identify an individual from this answer. Therefore, any queries answered in this trading framework should guarantee that the results do not reveal whether or not an individual is answering the query. Differential privacy can prevent the data linkage attacks on the published results of statistical/linear aggregated queries by introducing randomness. For these reasons, only statistical/linear aggregated queries should be allowed in the trading framework when the privacy is guaranteed by differential privacy. Li et al. (2017) also adopted this query type in their proposed theoretical framework.

Unlike those in references (Halevy, 2001; Nash et al., 2010), which studied the deterministic queries and views in depth, the query answers in our pricing method are random variables. Thus, the answerability is formally defined below.

Definition 9 (Answerability Li et al. (2017))**.** Let Q denote a query and let $S = \{Q_1, \ldots, Q_k\}$ denote a multi-set of queries; there exists a function $f : R^k \to R$, then any mechanism of $M_1, \ldots, M_k$, is able to answer the corresponding query $Q_1, \ldots, Q_k$, and there are $M(D) \stackrel{def}{=} f(M_1(D), \ldots, M_k(D))$ that can answer the query Q, thus, we can say that Q is answerable for S.

If f is linear, then Q is linear to S, which can be denoted as $S \to Q$. That is, S determines Q.

Proposition 1 (Li et al. (2017))**.** *Let $Q = (q, v)$ denote a query and let $S = \{(q_1, v_1), \ldots, (q_m, v_m)\}$ denote a multi-set of queries; then, the following two cases are equivalent.*

- *There exists $c_1, \ldots, c_m$ so that $c_1q_1 + \ldots + c_mq_m = q$ and $c_1^2v_1 + \ldots + c_m^2v_m \le v$.*
- $S \to Q$.

2.2.4. Arbitrage free

Before introducing the arbitrage-free pricing function, let us first look at the pricing function. We denote $\mathbb{R}^+ = [0, \infty)$ and $\bar{\mathbb{R}}^+ = \mathbb{R}^+ \cup \{\infty\}$.

Definition 10. A pricing function is $\pi : \mathbb{R}^n \times \bar{\mathbb{R}}^+ \to \bar{\mathbb{R}}^+$.

Let Q represent the buyer's query, which can be expressed as $Q = (q, v)$, where q represents the linear query, and v represents the quantity of noise that the buyer is willing to tolerate.Then the pricing function $\pi(Q) = \pi(q, v)$.

Arbitrage is the act of deducing a new query answer from a combination of multiple queries without paying the full price. A robust pricing function must ensure an arbitrage-free condition to prevent attackers from obtaining benefits through arbitrage.

Definition 11 (Arbitrage free Li et al. (2017))**.** Let $S = \{Q_1, \ldots, Q_m\}$ represent a multi-set of queries and let Q represent a query; if S determines Q, denoted as $S \to Q$, then the pricing function $\pi(Q)$ is arbitrage free, which can be expressed as formula Eqn 12:

$$\pi(Q) \le \Sigma_{i=1}^{m} \pi(Q_i) \tag{12}$$

If π allows arbitrage, then the buyer will arbitrage by buying the multi-set of queries that determines it.

Li et al. (2017) proved that any arbitrage-free pricing function must satisfy the following properties, as shown in Proposition 2.

Proposition 2 (Li et al. (2017))**.** *Let $\pi(q, v)$ denote an arbitrage-free pricing function; then,*

(1) If $q = 0$, then $\pi(0, v) = 0$.
(2) If $v' \ge v$, then $\pi(q, v') \le \pi(q, v)$.
(3) For $\forall v \ge 0$, $\pi(q, 0) \ge \pi(q, v)$ is true.
(4) If π is continuous at $q = 0$, then $\pi(q, \infty) = 0$.
(5) As $v \to \infty$, $\pi(q, v) = \Omega(1/v)$.

References Koutris et al. (2015); Li and Miklau (2012) studied the query answers in the deterministic context, and our pricing method extended it to make it a query answer with disturbance.

3. Arbitrage-free pricing function

Arbitrage free is a necessary economic property of the pricing function to prevent savvy data buyers from circumventing the query price. When the pricing function uses a fixed price, every query charges the same price; even zero queries charge the same price, which inevitably leads to arbitrage. When the cost of the pricing function is zero, i.e., every query is free, then there is no arbitrage. In this section, the sufficient conditions for the arbitrage-free pricing function $\pi(q, v)$ are given and we need to prove that there is an arbitrage-free pricing function.

We first analyze how the arbitrage-free pricing function $\pi(q, v)$ depends on the variance v. Then, according to item 2 of Proposition 2, $\pi(q, v)$ monotonically decreases in variance v, and according to item 4, it cannot be independent of the variance v. Proposition 3 shows that the arbitrage-free pricing function $\pi(q, v)$ cannot decrease faster than $1/v$.

Proposition 3. *Let $\pi(q, v)$ denote an arbitrage-free pricing function and let q denote a linear query; then, there is $\pi(q, v) = \Omega(1/v)$.*

Proof. If there is q and a sequence $\{v_i\}_1^\infty$ such that $\lim_{i\to\infty} v_i = \infty$ and $\lim_{i\to\infty} v_i\pi(q, v_i) = 0$, we select v_{i0} so that $v_{i0} > 1$ and $v_{i0}\pi(q, v_{i0}) < \pi(q, 1)/2$. Then, we can request the query $\pi(q, v_{i0})$ up to $\lceil v_{i0}\rceil$ times and calculate the average to answer $\pi(q, 1)$. The specific payment is as follows:

$$\lceil v_{i0}\rceil\pi(q, v_{i0}) \le (v_{i0} + 1)\pi(q, v_{i0}) < 2v_{i0}\pi(q, v_{i0}) < \pi(q, 1).$$

The existence of arbitrage behavior contradicts the fact that $\pi(q, v)$ is arbitrage free. This completes the proof.

Next, we prove that π is arbitrage free iff f is a semi-norm. If a function $f : \mathbb{R}^n \to \mathbb{R}^+$ satisfies the following two properties, it is called semi-norm.

- Homogeneity: $\forall c \in \mathbb{R}, q \in \mathbb{R}^n, f(cq) = |c| f(q)$.
- Subadditivity: $\forall q_1, q_2 \in \mathbb{R}^n, f(q_1 + q_2) \le f(q_1) + f(q_2)$.

If f is a semi-norm, then $q = 0$ exists and $f(q) = 0$. On the other hand, if $f(q) = 0$ exists and $q = 0$, then f is a norm. □

Theorem 5 (Basic Arbitrage-Free Pricing Function)**.** *For some function f, a pricing function π can be expressed as $\pi(q, v) = f^2(q)/v$, iff f is a semi-norm, and, thus, π is arbitrage free.*

According to Theorem 5, the arbitrage-free pricing function can be derived from the semi-norm function. Next, we will show how to combine the basic arbitrage-free pricing functions to construct more arbitrage-free pricing functions.

Theorem 6 (Composite Arbitrage-Free Pricing Function)**.** *Let $f : (\bar{\mathbb{R}}^+)^k \to \bar{\mathbb{R}}^+$ represent a function with a subadditivity and non-decreasing property. For any arbitrage-free pricing function $\pi_1, \ldots, \pi_k$, its composite pricing function $\pi = f(\pi_1, \ldots, \pi_k)$ is also arbitrage free. Here, $f(x + y) \le f(x) + f(y)$ means that the function is subadditive. If $f(x + y) \le f(x) + f(y)$, then $f(x) \le f(y))$, which means that the function is nondecreasing. According to the continuity and second-order differentiability of the function, it can be determined whether it is nondecreasing and subadditive.*

According to Theorems 5 and 6, more arbitrage-free pricing functions can be constructed. Some typical arbitrage-free pricing functions are given below.

Corollary 1 (Li et al. (2017))**.** *If $\pi_1, \ldots, \pi_k$ are arbitrage free, then all functions defined below are arbitrage free:*

- *L_2-norm:* $\pi(q, v) = \| q \|_2^2 / v = \Sigma_i q_i^2/v$.
- *L_∞-norm:* $\pi(q, v) = \| q \|_\infty^2 / v = \max_i q_i^2/v$.
- *Weighted L_2:* $\pi(q, v) = (\Sigma_i w_i \cdot q_i^2)/v$, *where* $w_1, \ldots, w_n \ge 0$.
- *Linear combination:* $c_1\pi_1 + \ldots + c_k\pi_k$, *where* $c_1, \ldots c_k \ge 0$.
- *Maximum:* $\max(\pi_1, \ldots \pi_k)$.
- *Cut-off:* $\min(\pi_1, c)$, *where* $c \ge 0$.
- *Power:* π_1^c, *where* $0 < c \le 1$.
- *Logarithmic:* $\log(\pi_1 + 1)$.
- *Geometric mean:* $\sqrt{\prod_{i=1}^k \pi_i}$.
- *Sigmoid:* $\arctan(\pi_1)$, $\tanh(\pi_1)$, $\pi_1/\sqrt{\pi_1^2 + 1}$.

The data market can select the corresponding arbitrage-free pricing functions according to different scenarios. For example, to limit the price of data to a limited range, one can use bounded pricing functions, such as cut-off functions and sigmoid functions.

4. Pricing framework

In this section, we consider the privacy compensation mechanism of personal big data. This section proposes two different pricing methods: positive pricing and reverse pricing from two different design ideas. In positive pricing, privacy compensation is first calculated according to the loss of privacy of each user, and then the query price is determined according to the total privacy compensation. In reverse pricing, the query price is calculated first and compensation is reasonably allocated according to the loss of privacy of the users. Positive pricing allows compensation to be made to individuals according to their privacy attitudes; however, the arbitrage-free property of the pricing function must be guaranteed.

4.1. *Privacy loss*

This section defines the privacy loss of personal big data owners based on differential privacy. Given two dependent neighboring datasets $D(L, R)$ and $D'(L, R)$, they differ only in one data item d_i and all other data items are identical. The privacy loss of the individual is measured by comparing the output results of random mechanism M on dependent neighboring datasets $D(L, R)$ and $D'(L, R)$ (Ghosh and Roth, 2015).

Definition 12 (Individual Privacy Loss)**.** Let M represent a random mechanism, and let $D(L, R)$ and $D'(L, R)$ represent two dependent neighboring datasets; then, the privacy loss of individual i is defined as formula (13):

$$\varepsilon_i(M) = \sup_{D(L,R),O} \mid log\frac{Pr(M(D(L, R)) \in O)}{Pr(M(D'(L, R)) \in O)} \mid \tag{13}$$

As can be seen from the definition of privacy loss of the individual, its size depends on the mechanism. We choose the Laplace mechanism to measure privacy loss, and privacy loss is the function output of variance and dependence sensitivity.

Theorem 7 (Niu et al. (2018))**.** *Let M represent a dependent perturbation mechanism v, the variance of Laplace noise; f, the query*

function; and DS_i^f, the dependent sensitivity. Then, the upper bound of the privacy loss of individual i can be expressed as formula (14):

$$\varepsilon_i(M) \leq \frac{DS_i^f}{\sqrt{v/2}} \tag{14}$$

4.2. Positive pricing

The principle that there cannot exist arbitrage opportunities in data markets is one of the most basic ideas of digital economics. An arbitrage is a portfolio that either is a strong arbitrage or has a positive, nonzero payoff and zero price. A strong arbitrage is a portfolio that has a positive payoff and a strictly negative price. The payoff pricing functional is linear and positive iff there is no strong arbitrage (Leroy and Werner, 2018).

Here, positive pricing means that the privacy compensation $\mu_i(Q)$ is calculated according to the privacy loss of each user, and then the query price $\pi(Q)$ is determined according to the total privacy compensation. Positive pricing allows individuals to actively select privacy compensation functions for their privacy attitude while ensuring an arbitrage-free condition.

Privacy loss $\varepsilon_i(M)$ decides the privacy compensation $\mu_i(Q)$, and the personal data bank should be able to evaluate $\varepsilon_i(M)$. Because the privacy loss $\varepsilon_i(M)$ defined in Definition 11 depends on the specific implementation mechanism, the entire dataset, and all possible outputs, the computational complexity is relatively high and impractical (Mcsherry and Talwar, 2007). The upper bound of privacy loss represented by Theorem 7 is a function of dependent sensitivity DS_i^f and variance v. Therefore, to calculate the privacy loss, we must calculate the dependent sensitivity first. Assuming that $\beta_i, \gamma_i \in \mathbb{R}$ represent the infimum and supremum of the value range of data item d_i, respectively, we can then calculate the dependence sensitivity by using formula (15):

$$DS_i^f = \sum_{j \in \mathbb{C}_i} \rho_{ij} |q_j| (\gamma_j - \beta_j) \tag{15}$$

Here, we assume that the data are independent of each other without considering the correlation between the data, namely, $\rho_{ij} = 0$. Then, formula (13) is simplified to the sensitivity defined in differential privacy (Dwork and Roth, 2014), as shown in formula (16):

$$DS_i^f = |q_i|(\gamma_j - \beta_j) \tag{16}$$

After quantifying the privacy loss of personal big data, we can consider how to calculate the privacy compensation. Here, we firstly define the fairness property that the privacy compensation need to satisfy.

Definition 13 (Fairness). Let $\mu_i(Q)$ denote a privacy compensation function over the query $Q = (q, v)$; for any $j \in \mathbb{C}_i$, if q_j=0, there is $\mu_i(Q) = 0$, then $\mu_i(Q)$ satisfy fairness.

Definition 14 (Positive Privacy Compensation). Let $\mu_i(Q)$ denote a privacy compensation function over the query $Q = (q, v)$ in the positive pricing. Then, $\mu_i(Q)$ satisfy these two properties:

- Fairness: $\forall j \in \mathbb{C}_i$, if q_j=0, then $\mu_i(Q)$=0.
- Arbitrage free: For each i, $\mu_i(Q)$ is arbitrage free.

Fairness means that the individual is compensated only if the individual data contribute to the query, i.e., if the individual data participate in the query. On the other hand, if the individual data did not participate in the query with any relevant data and did not contribute anything to the query, then the individual will not be compensated. Arbitrage free is an important economic attribute, which means that the buyer cannot avoid paying for the query by asking other queries or by combining the query answers to derive the desired results.

With reference to query pricing, we designed a generic positive pricing privacy compensation function to compensate individuals for their privacy loss. This compensation function is suitable for individuals who pay attention to privacy and do not want to accept full disclosure of privacy rights.

Theorem 8. *The generic privacy compensation functions can be expressed as formula Eqn 17:*

$$\mu_i(Q) = c_i \frac{DS_i^f}{\sqrt{v/2}} = c_i \frac{\sum_{j \in \mathbb{C}_i} \rho_{ij} |q_j| (\gamma_j - \beta_j)}{\sqrt{v/2}} \tag{17}$$

for some constants $c_i > 0$ and $\forall i \in \{1, \ldots, n\}$ are the basic positive privacy compensation functions.

Proof. First, we prove the fairness. We can prove it by $\forall j \in \mathbb{C}_i, q_j = 0 \Rightarrow \mu_i^1(Q) = \ldots = \mu_i^k(Q) = 0$. Next, we prove the arbitrage-free condition. Let $\mu_i(Q)$ be a linear combination of $\{|q_j|/\sqrt{v/2} | j \in \mathbb{C}_i\}$ and let $\{c_i \rho_{ij}(\gamma_j - \beta_j) | j \in \mathbb{C}_i\}$ be the corresponding coefficients; then By Corollary 1(linear combination:$c_1\pi_1 + \ldots + c_k\pi_k$), a linear combination $|q_j|/\sqrt{v/2}$ proves to be arbitrage free and by Corollary 1(geometric mean:$\sqrt{\prod_{i=1}^k \pi_i}$), a geometric mean $2q_j^2/v$ further proves to be arbitrage-free. Now, by using Corollary 1(weighted l_2 norm) and setting those weights, whose indexes are not j, to zeros, we complete our proof.

More positive privacy compensation functions can be constructed with reference to Corollary 1. For individuals who do not care much about privacy loss, privacy data can be disclosed at a limited price. In this section, we use the sigmoid function.

Theorem 9. *The tanh functions can be expressed as formula Eqn 18:*

$$\mu_i(Q) = b_i \tanh(c_i \frac{DS_i^f}{\sqrt{v/2}}) \tag{18}$$

for some constants $b_i, c_i > 0$ and $\forall i \in \{1, \ldots, n\}$ are bounded positive privacy compensation functions.

Proof. First, we prove the fairness by $\forall j \in \mathbb{C}_i, q_j = 0 \Rightarrow \mu_i^1(Q) = \ldots = \mu_i^k(Q) = 0$. Next, we have proved the arbitrage-free condition of $c_i DS_i^f/\sqrt{v/2}$ above. Thus, according to the sigmoid function and a linear combination of Corollary 1, it is arbitrage free. □

Theorem 10. *The arctan functions can be expressed as formula Eqn 19:*

$$\mu_i(Q) = b_i \arctan(c_i \frac{DS_i^f}{\sqrt{v/2}}) \tag{19}$$

for some constants $b_i, c_i > 0$ and $\forall i \in \{1, \ldots, n\}$ are bounded positive privacy compensation functions.

Proof. First, we prove the fairness by $\forall j \in \mathbb{C}_i, q_j = 0 \Rightarrow \mu_i^1(Q) = \ldots = \mu_i^k(Q) = 0$. Next, we have proved the arbitrage-free condition of *tanh* above. Thus, according to the sigmoid function of Corollary 1, it is arbitrage free.

Let $\mu_i(Q)$ be the arbitrage-free function, and then its linear combination $\pi(Q) = c\sum_{i=1}^n \mu_i(Q)$, $c > 1$ is also arbitrage free. Other comprehensive privacy compensation functions can also be obtained according to Corollary 1.

To sum up, when privacy compensation is fixed, positive pricing is more suitable for calculating privacy loss with the *tanh* function. Positive pricing allows individuals to actively select privacy compensation functions for their privacy attitude while ensuring an arbitrage-free condition. □

4.3. *Reverse pricing*

Information technology has made pricing strategies such as reverse pricing and revenue sharing ever easier to implement, especially when the transaction is done online. Correspondingly, Liu and Shuai (2015) expected such pricing strategies to become more relevant and their welfare implications more important over time.

Here, reverse pricing refers to the calculation of the query price $\pi(Q)$ according to the arbitrage-free pricing function in Section 3 and to the reasonable allocation of compensation according to the degree of privacy loss. Considering the cost factors of data collection, analysis, and mining, the personal data bank needs to keep part of the price $\pi(Q)$ paid by the buyer as the cost of data and then to allocate the rest to the individual users. Let $c\pi(Q)$ represent the privacy compensation allocated to the individual, where $0 < c < 1$. When $c = 0$, privacy compensation is 0; when $c = 1$, it means that all the payments paid by the buyer are compensated to the user. Therefore, either $c = 0$ or $c = 1$ is unreasonable. We can determine an appropriate compensation coefficient $c \in (0, 1)$ according to the specific data cost. Then, $(1 - c)\pi(Q)$ represents the data cost. Let $\mu_i(Q)$ be the privacy compensation function; then, $\sum_{i=1}^n \mu_i(Q) = c\pi(Q)$. The privacy compensation $\mu_i(Q)$ obtained by an individual shall be in direct proportion to its privacy loss $\varepsilon_i(M)$.

Because the dependent differential privacy proposed in Liu et al. (2016) is mainly applicable to general functions, its dependence sensitivity is relatively loose for specific environments. To limit the upper bound of the privacy loss function $\varepsilon_i(M)$, we need a tighter dependent sensitivity. We observed that the privacy loss of the individual is affected by the positive and the negative of the weight and the correlation. Therefore, let $\sigma_{ij} = 1$ represent the positive correlation of data items d_i and d_j, and let $\sigma_{ij} = -1$ represent the negative correlation of data items d_i and d_j, and let the original form of the weight be kept. The improved tight dependent sensitivity can be calculated by formula (20).

Lemma 1. *The tight dependent sensitivity of the data item d_i in the dependent dataset D to the query function $f = q^T D$ can be expressed as formula Eqn 20:*

$$DS_i^f = |\sum_{j \in \mathbb{C}_i} \sigma_{ij}\rho_{ij} q_j(\gamma_j - \beta_j)| \tag{20}$$

The privacy compensation function $\mu_i(Q)$ can be calculated by using the tight dependent sensitivity, as shown in formula (21).

Theorem 11. *The privacy compensation functions of reverse pricing are defined as formula Eqn 21:*

$$\begin{aligned}\mu_i(Q) &= c\pi(Q)\frac{DS_i^f/\sqrt{v/2}}{\sum_{i=1}^n DS_i^f/\sqrt{v/2}} \\ &= c\pi(Q)\frac{|\sum_{j\in\mathbb{C}_i}\sigma_{ij}\rho_{ij}|q_j|(\gamma_j-\beta_j)|}{\sum_{i=1}^n|\sum_{j\in\mathbb{C}_i}\sigma_{ij}\rho_{ij}|q_j|(\gamma_j-\beta_j)|}\end{aligned} \tag{21}$$

Here, $c\pi(Q)$ represents the sum of privacy compensation and $0 < c < 1$; DS_i^f represents the tight dependency sensitivity of the data item d_i. From formula (19), it can be seen that personal privacy compensation is in direct proportion to personal privacy loss.

Proof. The fairness of the compensation function is proved by $\forall j \in \mathbb{C}_i, q_j = 0 \Rightarrow \mu_i(Q) = 0$. Because the query price is first paid by the buyer, the buyer does not involve privacy compensation, and, therefore, he/she will not act as an attacker to obtain arbitrage. Thus, there is no need to guarantee an arbitrage-free condition in the privacy compensation function of reverse pricing.

To sum up, because the reverse pricing privacy compensation function does not need to guarantee that there will be no arbitrage, compared with the calculation of the positive pricing privacy compensation function, its calculation is more accurate and more suitable for linear aggregate statistics. □

5. Experimental analysis

We adopted Anaconda+Spyder as the experimental platform and implemented the pricing method in the Python programming language. We evaluated the performance of the pricing method (PMDP) on a real dataset. We priced and compensated individual data according to a privacy protection mechanism and an arbitrage-free pricing function (including positive pricing and reverse pricing).

We performed our experimental evaluation on the MovieLens 1 M dataset (MovieLens, 2003). This dataset contains 1,000,209 ratings of 3900 movies from 6040 anonymous users. The ratings in MovieLens were used as the target variables for supervised learning.

Personal profiles, such as gender, age, and occupation, were obtained as supporting data for calculating the dependent coefficients among the data items in accordance with the procedure described by Liu et al. (2016). According to Lemma 1, with $\sigma = 1$, the weight can be positive or negative. The individual's

Table 1 – Confusion matrix.

1	2	3	4	5
3738	2265	11,932	8881	1133
2448	2658	22,899	22,867	2695
2030	2790	42,458	72,843	10,798
890	1254	31,822	112,867	27,602
266	269	8907	64274	39,519

rating of the movie was taken as the target variable, and linear regression was applied to carry out a weighted sum and statistical learning under different dimensions and different weight vectors.

5.1. *Classification and effect evaluation*

The Movielens dataset scores a film on a scale of 1 to 5, with a higher score indicating a higher level of approval.The MovieID is the movie number, while the Rating is the response variable with a value ranging from 1 to 5. Logical regression method was used to divide the movie scores into five ratings, and scikit-learn was used to train the classifier. The confusion matrix is used to visualize the data of different classification errors, as shown in Table 1. The accuracy, precision, recall, and comprehensive evaluation index (F1-score) of each type and the overall category index are shown in Table 2. The classification accuracy from rating 1 to rating 5 was 0.94, 0.89, 0.67, 0.54 and 0.77, respectively, and the overall accuracy reached 0.76, which achieved a very good classification effect. We obtained the best parameter combination through grid search, and the final classifier was obtained through continuous optimization of the original classifier.

5.2. *Privacy and utility*

This section evaluates how our pricing method (PMDP) can improve accuracy, better protect data privacy, and improve the utility of data aggregation statistics compared with differential privacy (DP) and dependent differential privacy (DDP). Our pricing method allows for a more accurate calculation of dependency sensitivity by using the mechanism defined in Theorem 4. We selected the ratings of 1000 users about the movie to train, to obtain a 1000-dimensional weight vector. Let $1-\frac{\|f(D)-f'(D)\|_1}{\|f(D)+f'(D)\|_1}$ (Liu et al., 2016) represent the accuracy of the calculation, and let $f(D)$ and $f'(D)$ respectively represent the real output and the output with noise interference. Fig. 2 compares the accuracy of the weighted sum statistics of DP, DDP, and PMDP under a Laplace mechanism (Dwork and Roth, 2014).

As can be seen from Fig. 2, privacy budget ϵ is in direct proportion to accuracy. The larger the value of ϵ, the higher the accuracy. When the privacy budget is a fixed value, PMDP is more accurate than DDP or even than DP. When $\epsilon = 0.1$, PMDP improved the accuracy by 32.66% compared with DP and by 24.3% compared with DDP. When ϵ is too small, the interference results will dominate, whereas when it is too large, the real results will dominate. In these two cases, the difference in accuracy among the three methods becomes very small.

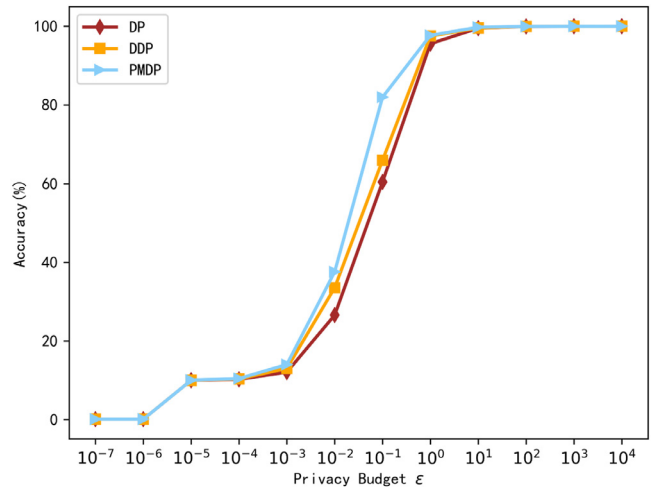


Fig. 2 – Comparison of the accuracy of the weighted sum statistics of DP, DDP, and PMDP under a Laplace mechanism.

To sum up, PMDP has a higher accuracy than those of DP and DDP, and can better protect individual data privacy through differential privacy and improve data utility.

5.3. *Pricing function*

This section evaluates the pricing function in terms of noise variance, weighted sum, and arbitrage free.

5.3.1. *Noise variance*

Through a simulation attack, the effect of noise variance v on arbitrage-free pricing function was evaluated, and the defense degree of different variances against arbitrage behavior was analyzed. Using the 10,000 data samples to simulate the attack behavior, we plotted the cumulative fraction of the ratio of attack cost $\sum_{j=1}^{m}\pi(q,v_j)$ to the original price $\pi(q,v)$. As can be seen from Fig. 3, price $\pi(\cdot)$ gradually decreases with variance v from $1/v^2$ to $1/v$ and then to $1/\sqrt{v}$. The cumulative fraction with a ratio of 1 is the critical point of discovering the arbitrage behavior, reflecting the success rate of discovering such behavior.

As can be seen from the cumulative fraction in Fig. 3, when the ratio of attack cost to the original price is 1, arbitrage is free when the variance is $1/v$ and $1/\sqrt{v}$, whereas arbitrage exists when the variance is $1/v^2$. In addition, it can be seen from the area graph of the function curve of the cumulative fraction that the attack cost of arbitrage behavior is higher when the variance is $1/\sqrt{v}$ than when the variance is $1/v$. Thus, as the variance v decreases even more slowly, the pricing function will become increasingly more resistant to arbitrage.

5.3.2. *Weight vector*

We discuss the pricing problem of the L_p norm and *tanh* function under different dimensions of the weight vector. The weight vector n has dimensions from 1000 to 5000 with steps of 500. From L_1, L_2, L_3, L_∞, four norms plus the *tanh* function form the four comprehensive pricing functions. As can be seen from Fig. 4, price is in direct proportion to the dimension of the

Table 2 – Classification report.

Category	TP	FN	FP	TN	Accuracy	Precision	Recall	F1-score
1	3738	24,211	5634	466,522	0.94	0.40	0.13	0.20
2	2658	50,909	6578	439,960	0.89	0.29	0.05	0.08
3	42,458	88,461	75,560	293,626	0.67	0.36	0.32	0.34
4	112,867	61,568	168,865	156,805	0.54	0.40	0.65	0.49
5	39,519	73,716	42,228	344,642	0.77	0.48	0.35	0.41
Total	201,240	298,865	298,865	1,701,555	0.76	0.40	0.40	0.40

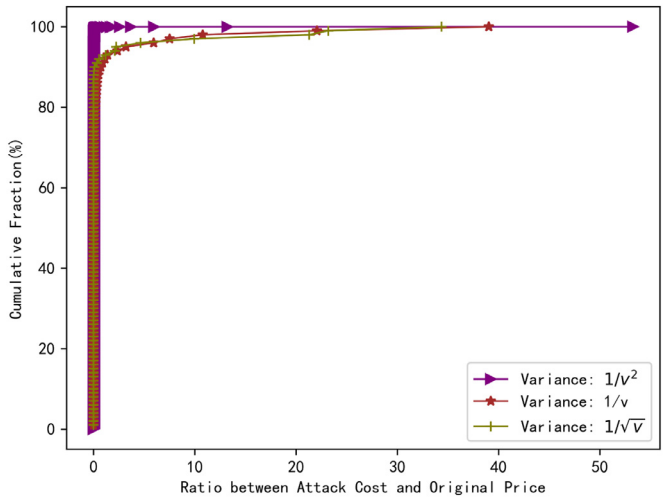


Fig. 3 – Plot of the noise variance.

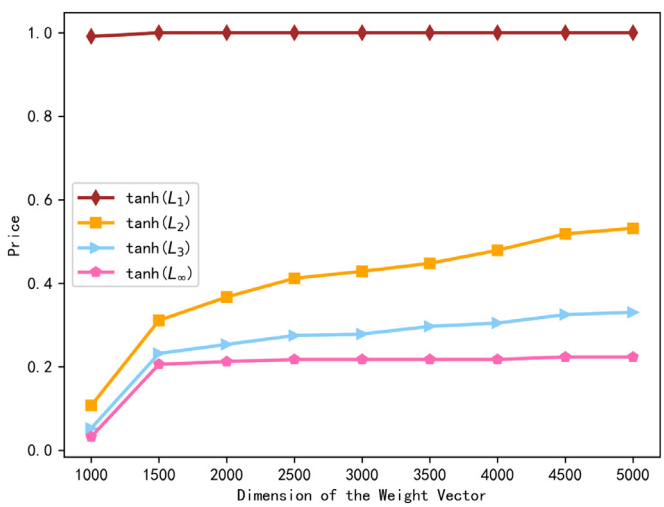


Fig. 4 – Plot of the L_p norm and *tanh*.

weight vector. For a certain pricing function, the larger the dimension of the weight vector is, the higher the price will be.

It can be seen from Fig. 4 that the price of the L_1 pricing function is higher than that of the one for L_2, the price of the L_2 pricing function is higher than that of the one for L_3, and the price of the L_3 pricing function is higher than that of the one for L_∞. In addition, we can see that the price of the L_1 pricing function basically remains at 1 because, when the dimension reaches 3000, the price of the L_1 pricing function is close to the upper bound of the *tanh* function.

In conclusion, when the weight vector dimension is fixed, the price of the L_p pricing function is inversely proportional to p, and the higher the value of p, the lower the price. For the same type of pricing function, the price has a proportional relationship to the dimension of the weight vector: the larger the dimension of the weight vector, the higher the price.

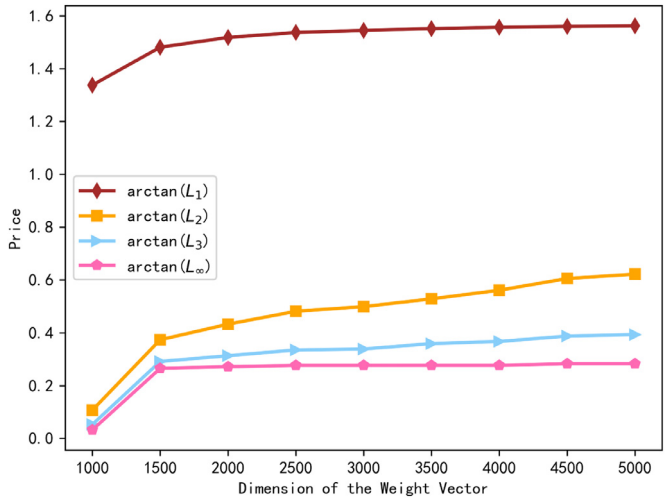


Fig. 5 – Plot of the L_p norm and *arctan*.

Similarly, according to Corollary 1, the *arctan* function is also an arbitrage-free function. The weight vector n has dimensions from 1000 to 5000 with steps of 500. From L_1, L_2, L_3, L_∞, four norms plus the *arctan* function form the four comprehensive pricing functions. As can be seen from Fig. 5, price is also in direct proportion to the dimension of the weight vector. For a certain pricing function, the larger the dimension of the weight vector is, the higher the price will be.

It can be seen from Fig. 5 that the price of the L_1 pricing function is higher than that of the L_2 one, the price of the L_2 pricing function is higher than that of the L_3 one, and the price of the L_3 pricing function is higher than that of the L_∞ one. In addition, we can see that the price of the L_1 pricing function basically remains at 1.57 because, when the dimension reaches 3000, the price of this function is also close to the upper bound of the *arctan* function.

Similarly, when the weight vector dimension is fixed, the price of the L_p pricing function is inversely proportional to p, and the higher the value of p, the lower the price. For the same type of pricing function, the dimension of the price and weight vector is also proportional: the larger the dimension of the weight vector is, the higher the price is.

5.3.3. *Arbitrage-free pricing*

The smaller the variance v, the better the privacy protection, the lower the privacy loss of the data owner, the better the data authenticity, and the higher the corresponding price. In contrast, the greater the variance v, the worse the privacy protection, the greater the privacy loss of the data owner, the greater the data interference, and the lower the corresponding price. We discuss the effect of an arbitrage-free pricing function from two dimensions of the weight vector and variance.

The dimension of the weighting vector varies from 1000 to 5000, the step length is 500, and the variance is 0.1, 1, 2, 3, 4, 5, and 10. Seven different arbitrage-free pricing functions are constituted by the L_1 norm, *tanh* function, and seven different values of variance v.

As can be seen from Fig. 6, when the dimension n of the weight vector is fixed, the larger the variance v, the lower the price. This further proves that the arbitrage-free pricing function is inversely proportional to the variance v: the larger the dimension n of the weight vector of the same arbitrage-free pricing function is, the higher the price will be. The arbitrage-free pricing function is in direct proportion to the dimension n of the weight vector.

To sum up, it is summarized as follows through Table 3.

Table 3 – Influence of noise variance, weight Vector and L_p on pricing function.

Pricing function	Noise variance	Weight vector	L_p norm
$\pi(q, v)$	Inverse proportion	Direct proportion	Inverse proportion
tanh	Inverse proportion	Direct proportion	Inverse proportion
arctan	Inverse proportion	Direct proportion	Inverse proportion

5.4. *Privacy compensation*

This section discusses how to make privacy compensation under aggregate statistics. For the convenience of comparison and illustration, we fixed the total compensation to 10,000 to compensate 1000 users, and each data owner was compensated 10 units on average. We select the privacy compensation function of Theorem 9 for the positive pricing and the privacy compensation function of Theorem 11 for the reverse pricing to provide a fair and reasonable privacy compensation for each data owner.

We perform a weighted sum, where the dimension of the weight vector is 1000 and the variance v is 0.1. Fig. 7 plots the positive pricing privacy compensations under DP. Fig. 8 plots the positive pricing privacy compensations under DDP. Fig. 9 plots the positive pricing privacy compensations under PMDP. As can be seen from Figs. 7–9, most data owners are evenly and reasonably compensated, whereas only a few data owners

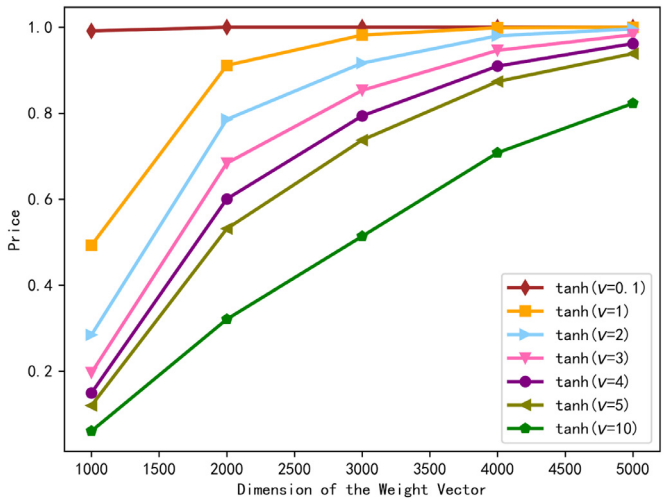


Fig. 6 – Price comparisons under different variances.

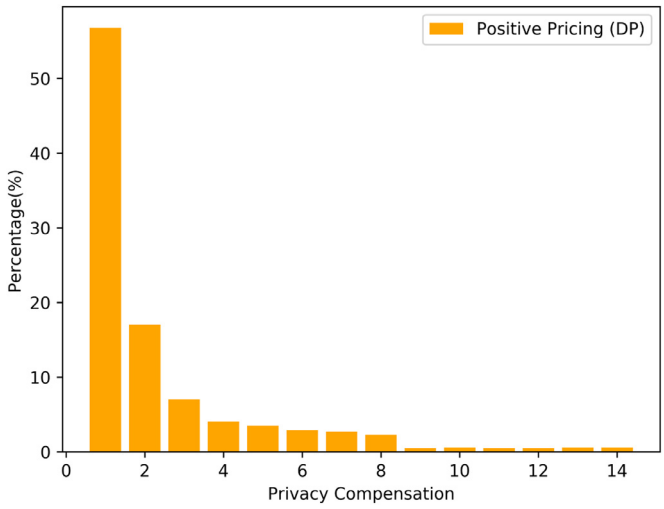


Fig. 7 – Chart of the positive pricing privacy compensation based on differential privacy (DP).

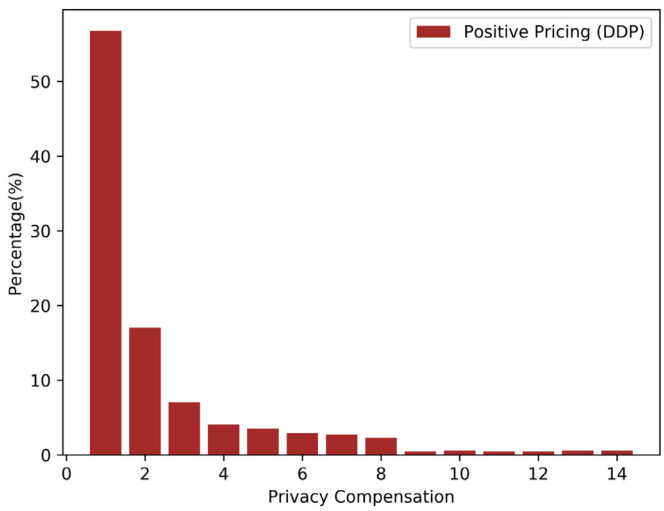


Fig. 8 – Chart of the positive pricing privacy compensation based on dependent differential privacy (DDP).

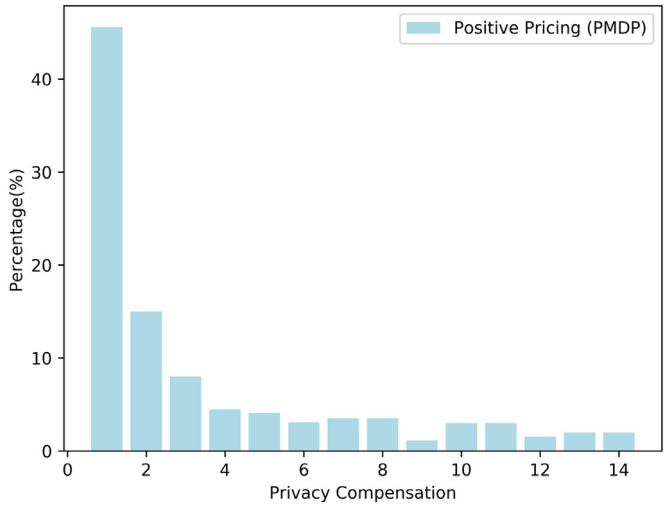


Fig. 9 – Chart of the positive pricing privacy compensation based on PMDP.

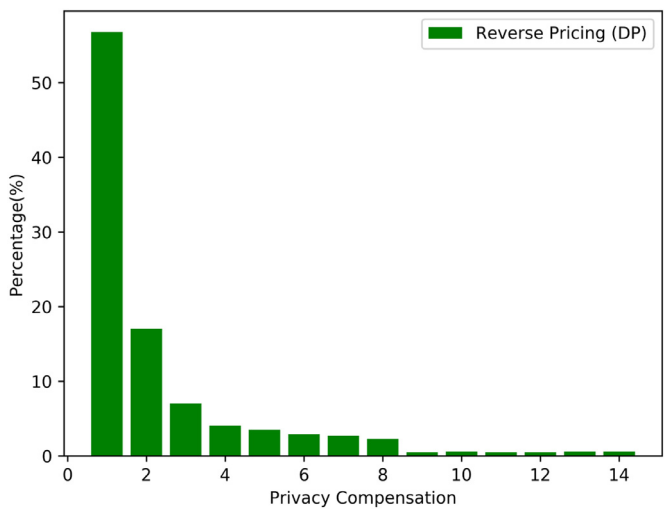


Fig. 10 – Chart of the reverse pricing privacy compensation based on differential privacy (DP).

receive more compensation. This is basically consistent with the distribution phenomenon in a real-world setting. It can be seen from the figures that the three types of positive pricing compensation based on DP, DDP, and PMDP are basically consistent.

Fig. 10 plots the reverse pricing privacy compensations under DP. Fig. 11 plots the reverse pricing privacy compensations under DDP. Fig. 12 plots the reverse pricing privacy compensations under PMDP. As can be seen from Figs. 10–12, reverse pricing compensation is slightly different from positive pricing compensation. Reverse pricing compensation looks more reasonable. We first compare DP and DDP in reverse pricing privacy compensation. As can be seen from Figs. 10 and 11, more privacy compensations fall into the middle of the DDP compared with DP. It further indicates that DDP has more data owners who obtain privacy compensation when considering data dependency. Similarly, it can be seen from Figs. 11 and 12 that, compared with DDP, PMDP has more privacy compensations falling into the intermediate area, namely, PMDP has more data owners who receive privacy compensation. Thus, compared with DP and DDP, PMDP is more fair and reasonable, and considers more the dependencies between data, so that more data owners can obtain privacy compensation. It is summarized as follows through Table 4.

To sum up, whether it is positive pricing or reverse pricing, individual privacy compensation is in direct proportion to the degree of privacy loss. In addition, it is more suitable for positive pricing to use the *tanh* function to calculate privacy loss, whereas reverse pricing is more accurate. It is summarized as follows through Table 5.

6. Related work

This section introduces relevant works in terms of data pricing mechanism design, query-based pricing, auction-based pricing, and differential privacy-based pricing.

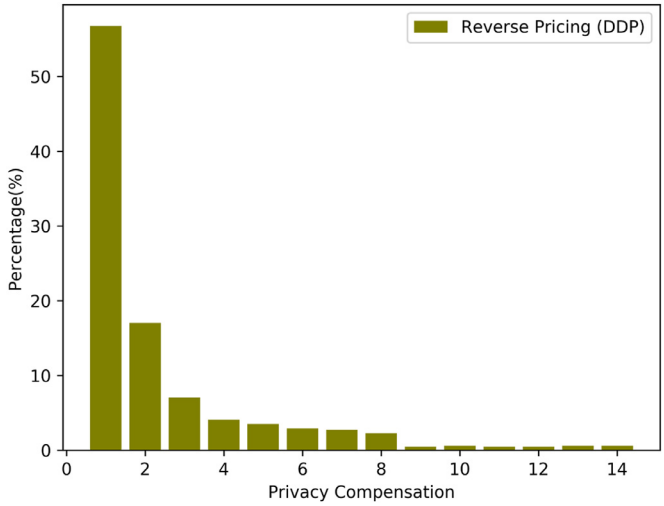


Fig. 11 – Chart of the reverse pricing privacy compensation based on dependent differential privacy (DDP).

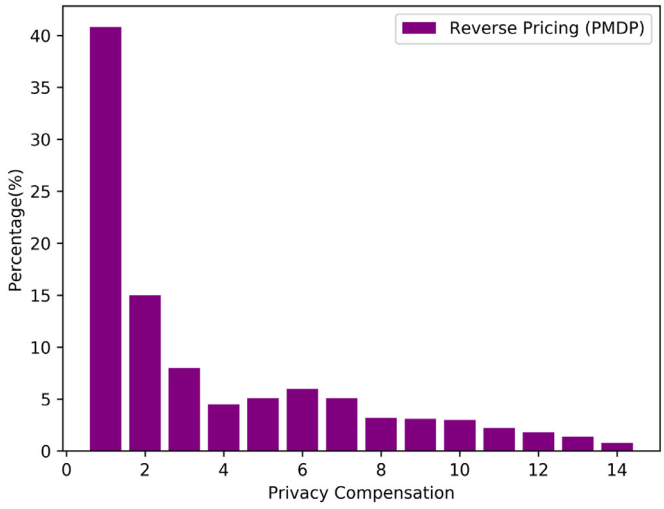


Fig. 12 – Chart of the reverse pricing privacy compensation based on PMDP.

Table 4 – Comparison of the privacy compensations of DP, DDP, and PMDP.

Pricing method	Accuracy	Fairness	Distribution	Number of individuals receiving compensation	Data correlation
PMDP	More	Strong	Uniform	More	Strong
DDP	Lower	Weak	Discrete	Less	Strong
DP	Lower	Weak	Discrete	Less	Weak

Table 5 – Comparison of positive pricing and reverse pricing.

Pricing mechanism	Arbitrage-free	Accuracy	Privacy loss	Applicability	Individual selectivity
Positive pricing	Need	Lower accurate	Direct proportion	*tanh* function	Strong
Reverse pricing	Does not need	More accurate	Direct proportion	Linear aggregate statistics	Weak

Data pricing method design Fricker and Maksimov (2017) provided an overview of pricing methods for data markets, summarizing some of the attributes that need to be considered in pricing, such as the bid, data usage, data quality, price of the view or tuple, and cost. Steppe (2017) provided a theory on whether, and how, the General Data Protection Regulation(GDPR) affects price discrimination. Bataineh et al. (2016) proposed a trading platform that allows individuals to trade and monetize their data. Kiyomoto (2016) suggested that an optimal solution can be found between value and anonymity when publishing personal big data. Liang et al. (2018) focused on designing a data trading platform that can achieve efficiency, security, and privacy protection. Chen et al. (2017) proposed a practical framework for pricing cloud datasets and resources. Nget et al. (2018) proposed a practical personal big data trading framework that strikes a balance between money and privacy. In the differentially private top k selection problem, Bafna and Ullman (2017) considered that this can be solved with the well-known exponential mechanism and composition properties of differential privacy. Bun et al. (2016) considered that the query of a sensitive dataset is limited by differential privacy. Yu and Zhang (2017) suggested that data quality and data version strategy should be considered in data pricing. Codd (1983) argued that the users of a data bank must be protected and should be informed of how the data are organized in the computer. Stahl and Vossen (2013) believed that accurate data services can be priced according to data usage. Various data mining algorithms for personal privacy protection are classified and extensibility described in Verykios et al. (2004). Mao et al. (2019) studied how to maximize the benefits of the Internet-of-Things data market. Alelyani et al. (2017) proposed a context-centric software crowdsourcing pricing method.

Query-based pricing Koutris et al. (2012) presented a query market that can automatically price ad hoc queries. Zheng et al., 2019 proposed a crowd perception data pricing mechanism based on online queries. Shen et al., 2019 proposed a personal big data pricing method based on minimum provenance. The query-based pricing method can guarantee the two important properties of no arbitrage and no discount, but it is mainly applied to the traditional structured data without consideration of the protection of data privacy.

Auction-based pricing GGhosh and Roth (2015) studied how to motivate individuals to contribute their privacy data and suggested auction pricing for single-count queries. Niu et al. (2019b) provided a security scheme for data markets that can guarantee data authenticity and privacy protection at the same time. The goal is to devise a mechanism for disclosing true valuations. Wang et al. (2019b) proposed a method of auction query that provides the possibility of using a non-fixed pricing model to trade queries. Zhu et al. (2014) proposed a differential privacy spectrum auction mechanism to maximize revenue. The auction-based pricing allows buyers and sellers to negotiate an agreement, but the data is either fully disclosed or fully protected, without consideration for privacy assessments.

Differential privacy-based pricing Differential privacy can improve data utility while protecting privacy. There are a large number of studies in the literature on differential privacy, the most important one of which is Dwork and Roth (2014). Dinur and Nissim (2003) initiated a survey of the trade-offs between data privacy and data utility in a dataset, and Dwork et al. introduced differential privacy and a Laplace mechanism (Dwork et al., 2017). Zhang et al. (2011) proposed how to use differential privacy to limit the disclosure of personal information in data mining. Malgieri and Custers (2018) analyzed whether individuals have the right to know the value of their personal data. Abadi et al. (2016) believed that, when publishing a statistical query, privacy loss exists from the perspective of differential privacy. Chao et al. (2014); Li et al. (2017) proposed a private data pricing theory that realizes data pricing through differential privacy and a Laplace mechanism and that makes reasonable compensation for personal privacy loss. Gkatzelis et al. (2015) proposed a data market in which purchasers can obtain unbiased samples of private data according to their privacy attitudes. Zhu et al. (2017) conducted a multidisciplinary differential privacy survey for data publishing and data analysis. Niu et al. (2018) proposed a data market pricing framework that can perform common aggregate statistics for privacy correlated data. Task and Clifton (2012) proposed a new differential privacy standard for network data, i.e., out-link privacy. Wang et al. (2019a) proposed a differential evolution method based on a support vector machine classifier to forecast price classification. Zhang et al. (2012) proposed a differential privacy method for large optimization analysis. Niu et al. (2017) proposed a mechanism called Truthfulness and Privacy preservation in Data Markets (TPDM) that can effectively combine data truthfulness and privacy protection in the data market. Stahl and Vossen (2016) discussed data quality and proposed a weighted quality score to allow a comparison of the "cost performance" of different products. Parra-Arnau (2018) proposed a method for solving monetary privacy exchange and described the trade-off between information disclosure and economic return. Xu et al. (2017) studied the privacy evaluation of multiple users randomly selected from an unknown distribution for pricing. Yang et al. (2015) argued that differential privacy can be vulnerable when the data are relevant and the attacker does not have full background knowledge. Song et al. (2017) studied how to use Pufferfish to protect data privacy. Niu et al. (2019a) employed Pufferfish privacy to quantify the privacy loss and make a reasonable compensation for individuals. The differential privacy-based pricing can compensate for the loss of privacy, but adding noise will affect data quality and reduce data practicability.

Data quality is an important factor affecting the price; accurate and non-perturbation data should have a higher price and vice versa. However, to protect privacy, we need to add noise to the true data. Finding a balance between privacy and utility is what we are looking for. The original aim of our research work was to protect and price personal big data reasonably through differential privacy. In this paper, based on the investigation of the existing pricing methods, combined with the characteristics of personal big data, using the query-based pricing method; Considering data security and data quality, differential privacy method is adopted to protect data. Based on the existing differential privacy and dependent differential privacy, PMDP pricing method is proposed. Our pricing method can not only protect data privacy, but also realize data utiliza-

tion, and make reasonable compensation for the privacy loss of data owners.

7. Discussion

The method that we proposed adopts noise disturbance mechanism to provide strong differential privacy protection for personal data, which is compatible with GDPR Article 6-Lawfulness of processing and Article 32-Security of processing. Our aim is to protect personal data that is consistent and compatible with GDPR. At the same time, our method can pricing data on the basis of fully protecting the security of personal data, promote the transaction and sharing of personal data, and achieve a dynamic balance between open sharing and privacy protection.

Our method can make appropriate compensation according to the degree of privacy loss through two mechanisms, namely positive and reverse, so as to protect data privacy. When the privacy compensation is fixed, the positive mechanism is more suitable to calculate the privacy loss. The reverse mechanism is more suitable for linear aggregation statistics, and the calculation of privacy loss is more accurate, and the privacy compensation is more reasonable. Through the positive mechanism and the reverse mechanism, privacy compensation can be made fairly and reasonably, and the rationality and fairness of privacy budget allocation can be guaranteed.

Our method can provide effective privacy protection and data pricing for private data in different application scenarios, such as movie scores, recommendation systems, social networks, search logs and medical data.

In this paper, we discuss the numerical data pricing method based on differential privacy and use Laplace mechanism to add noise to calculate differential privacy. However, there are a lot of non-numerical data in the practical application, so how to conduct pricing research on non-numerical data will be a very difficult problem. For non-numerical data, the exponential mechanism and scoring function can be used to calculate the differential privacy. Therefore, using exponential mechanism to realize non-numerical data pricing based on differential privacy will be the next research direction.

8. Conclusion

The pricing method (PMDP) proposed in this paper can be used for personal big data pricing and privacy loss compensation. First, PMDP can make different pricing according to different variances. The smaller the variance, the better the privacy protection, and the lower the price. Next, PMDP can compensate individuals for their privacy losses using both positive pricing and reverse pricing. When privacy compensation is fixed, positive pricing is more suitable for calculating privacy loss with the *tanh* function. Reverse pricing does not need to guarantee that there will be no arbitrage and is more suitable for linear aggregate statistics, and the privacy loss calculation is more accurate and the corresponding privacy compensation is distinct. We perform aggregate statistics on a real dataset and evaluated their performance extensively. The experimental results show that PMDP can fairly and reasonably provide fine-grained privacy compensation to data owners through positive pricing and reverse pricing, ensure the practicability of the data, and achieve a balance between privacy protection and data utility. In addition, PMDP has a higher accuracy than those of DP and DDP, and can better protect individual data privacy through differential privacy and improve data utility.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yuncheng Shen: Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Bing Guo:** Funding acquisition, Methodology, Project administration, Supervision. **Yan Shen:** Software. **Xuliang Duan:** Resources. **Xiangqian Dong:** Investigation. **Hong Zhang:** Data curation. **Chuanwu Zhang:** Visualization. **Yuming Jiang:** Validation.

REFERENCES

Abadi M, Chu A, Goodfellow I, Mcmahan H, Mironov I, Talwar K, Zhang L. Deep learning with differential privacy. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security. ACM; 2016. p. 308–18.

Alelyani T, Mao K, Yang Y. Context-centric pricing: farly pricing models for software crowdsourcing tasks. In: Proceedings of the 13th International Conference on Predictive Models and Data Analytics in Software Engineering. ACM; 2017. p. 63–72.

Backstrom L, Sun E, Marlow C. Find me if you can: improving geographical prediction with social and spatial proximity. In: Proceedings of the 19th International Conference on World Wide Web. ACM; 2010. p. 61–70.

Bafna, M., Ullman, J., 2017. The price of selection in differential privacy. arXiv.org URL: http://search.proquest.com/docview/2074674569/.

Bataineh AS, Mizouni R, Barachi ME, Bentahar J. Monetizing personal data: a two-sided market approach. Procedia Comput. Sci. 2016;83:472–9.

Bun, M., Steinke, T., Ullman, J., 2016. Make up your mind: the price of online queries in differential privacy. arXiv.org URL: http://search.proquest.com/docview/2077033886/.

Chao L, Daniel Yang L, Gerome M, Dan S. A theory of pricing private data. ACM Trans. Database Syst. 2014;39(4):1–28. doi:10.1145/2691190.2691191.

Chen L, Koutris P, Kumar A. Model-based pricing: do not pay for more than what you learn. In: Proceedings of the 1st Workshop on Data Management for End-to-end Machine Learning. ACM; 2017. p. 1–4.

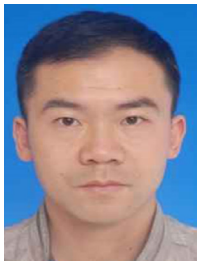
Codd E. A relational model of data for large shared data banks. Commun. ACM 1983;26(1):64–9.

Cummings R, Ligett K, Roth A, Wu ZS, Ziani J. Accuracy for sale: aggregating data with a variance constraint. In: Proceedings of the 2015 Conference on Innovations in Theoretical Computer Science. ACM; 2015. p. 317–24.

Dinur I, Nissim K. Revealing information while preserving privacy. In: Proceedings of the Twenty-second ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems. ACM; 2003. p. 202–10.

Dwork C, Mcsherry F, Nissim K, Smith A. Calibrating noise to sensitivity in private data analysis. J. Privacy Confid. 2017;7(3):17–51.
Dwork C, Roth A. The algorithmic foundations of differential privacy. Found. Trends Theor. Comput. Sci. 2014;9(3-4):211–407.
Facebook, 2018. Facebook is not the problem. lax privacy rules are.https://www.nytimes.com/2018/04/01/opinion/facebooklax-privacy-rules.html/.
Fricker, S., Maksimov, Y., 2017. Pricing of data products in data marketplaces. vol. 304, pp. 49–66.
Ganzhao Y, Zhenjie Z, Marianne W, Xiaokui X, Yin Y, Zhifeng H. Optimizing batch linear queries under exact and approximate differential privacy. ACM Trans. Database Syst. (TODS) 2015;040(002):1–47.
Ghosh A, Roth A. Selling privacy at auction. Games Econ. Behav. 2015;91(C):334–46.
Gkatzelis V, Aperjis C, Huberman B. Pricing private data. Electron. Mark. 2015;25(2):109–23.
Guo B, Qiang LI, Duan XL, Shen YC, Dong XQ, Zhang H, Shen Y, Zhang ZL, Luo J. Personal data bank:a new mode of personal big data asset management and value-added services based on bank architecture. Chin. J. Comput. 2017;40(1):126–43.
Haddadi H, Mortier R, Hand S. Privacy analytics. Acm Sigcomm Comput. Commun. Rev. 2012;42(2):94–8.
Halevy AY. Answering queries using views: a survey. VLDB J. 2001;10(4):270–94.
Jorgensen Z, Yu T, Cormode G. Conservative or liberal? Personalized differential privacy. In: 2015 IEEE 31st International Conference on Data Engineering. IEEE; 2015. p. 1023–34.
Kiyomoto, S., 2016. Data value estimation for privacy-preserving big/personal data businesses.
Koutris P, Upadhyaya P, Balazinska M, Howe B, Dan S. Query-based data pricing. J. ACM 2015;62(5):1–44.
Koutris P, Upadhyaya P, Balazinska M, Howe B, Suciu D. Querymarket demonstration: pricing for online data markets. Proc. VLDB Endow. 2012;5(12):1962–5.
Leroy, S., Werner, J., 2018. Arbitrage and Positive Pricing. pp. 24–34. 10.1017/cbo9781139162272.005
Li C, Li D, Miklau G, Suciu D. A theory of pricing private data. Commun. ACM 2017;60(12):79–86.
Li, C., Miklau, G., 2012. Pricing aggregate queries in a data marketplace. Webdb.
Liang F, Yu W, An D, Yang Q, Fu X, Zhao W. A survey on big data market: pricing, trading and protection. Ieee Access 2018;6:15132–54.
Liu CC, Chakraborty S, Mittal P. In: The Network and Distributed System Security Symposium 2016. Dependence makes you vulnerable: differential privacy under dependent tuples. Internet Society; 2016.
Liu Q, Shuai J. Reverse pricing and revenue sharing in a vertical market. Manag. Decis. Econ. 2015;36(5):299–313.
Malgieri G, Custers B. Pricing privacy – the right to know the value of your personal data. Comput. Law Secur. Rev. 2018;34(2):289–303.
Mao W, Zheng Z, Wu F. Pricing for revenue maximization in IoT data markets: an information design perspective. In: IEEE INFOCOM 2019 - IEEE Conference on Computer Communications. IEEE; 2019. p. 1837–45.
Mcsherry F, Talwar K. Mechanism design via differential privacy. In: 48th Annual IEEE Symposium on Foundations of Computer Science (FOCS'07). IEEE; 2007. p. 94–103.
MovieLens, 2003. Movielens 1mdataset. 2003. Available online. https://grouplens.org/datasets/movielens/1m/.
Nash A, Segoufin L, Vianu V. Views and queries: determinacy and rewriting. ACM Trans. Database Syst. (TODS) 2010;35(3):1–41.
Nget, R., Cao, Y., Yoshikawa, M., 2018. How to balance privacy and money through pricing mechanism in personal data market. arXiv.org URL: http://search.proquest.com/docview/2068911597/.
Niu C, Zheng Z, Tang S, Gao X, Wu F. Making big money from small sensors: trading time-series data under pufferfish privacy. In: IEEE INFOCOM 2019 - IEEE Conference on Computer Communications. IEEE; 2019. p. 568–76.
Niu C, Zheng Z, Wu F, Gao X, Chen G. Trading data in good faith: integrating truthfulness and privacy preservation in data markets. In: 2017 IEEE 33rd International Conference on Data Engineering (ICDE). IEEE; 2017. p. 223–6.
Niu C, Zheng Z, Wu F, Gao X, Chen G. Achieving data truthfulness and privacy preservation in data markets. IEEE Trans. Knowl. Data Eng. 2019;31(1):105–19.
Niu C, Zheng Z, Wu F, Tang S, Gao X, Chen G. Unlocking the value of privacy: trading aggregate statistics over private correlated data. In: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM; 2018. p. 2031–40.
Parameswaran K, Victor P, Hassan Jameel A, Mohamed Ali K. Not all attributes are created equal: d_x-private mechanisms for linear queries. Proc. Privacy Enhancing Technol. 2020;2020(001):103–25.
Parra-Arnau J. Optimized, direct sale of privacy in personal data marketplaces. Inf. Sci. 2018;424(C):354–84.
Shen Y, Guo B, Shen Y, Duan X, Dong X, Zhang H. A pricing model for big personal data. Tsinghua Sci. Technol. 2016;21(5):482–90.
Shen Y, Guo B, Shen Y, Wu F, Zhang H, Duan X, Dong X. Pricing personal data based on data provenance. Appl. Sci. 2019;9(16):1–21.
Song S, Wang Y, Chaudhuri K. Pufferfish privacy mechanisms for correlated data. In: Proceedings of the 2017 ACM International Conference on Management of Data. ACM; 2017. p. 1291–306.
Stahl F, Vossen G. High quality information provisioning and data pricing. In: 2013 IEEE 29th International Conference on Data Engineering Workshops (ICDEW). IEEE; 2013. p. 290–3.
Stahl F, Vossen G. Data Quality Scores for Pricing on Data Marketplaces. Springer Verlag; 2016. p. 215–24.
Steppe R. Online price discrimination and personal data: ageneral data protection regulation perspective. Comput. Law Secur. Rev. 2017;33(6):768–85.
Task C, Clifton C. A guide to differential privacy theory in social network analysis. In: 2012 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining. IEEE; 2012. p. 411–17.
Twitter, 2018. Twitter sold data access to cambridge analytica-linked researcher. https://www.bloomberg.com/news/articles/2018-04-29/twitter-sold-cambridge-analytica-researcher-public-dataaccess/.
Verykios V, Bertino E, Fovino I, Provenza L, Saygin Y, Theodoridis Y. State-of-the-art in privacy preserving data mining. ACM SIGMOD Rec. 2004;33(1):50–7.
Wang K, Xu C, Zhang Y, Guo S, Zomaya AY. Robust big data analytics for electricity price forecasting in the smart grid. IEEE Trans. Big Data 2019;5(1):34–45.
Wang X, Wei X, Gao S, Liu Y, Li Z. A novel auction-based query pricing schema. Int. J. Parallel Program. 2019;47(4):759–80.
Xiaoguang L, Hui L, Hui Z, Muyang H. The optimal upper bound of the number of queries for laplace mechanism under differential privacy. Inf. Sci. 2019;503(001):219–37.
Xu L, Jiang C, Qian Y, Zhao Y, Li J, Ren Y. Dynamic privacy pricing: a multi-armed bandit approach with time-variant rewards. IEEE Trans. Inf. Forensics Secur. 2017;12(2):271–85.
Yang B, Sato I, Nakagawa H. Bayesian differential privacy on correlated data. In: Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data. ACM; 2015. p. 747–62.

Yu H, Zhang M. Data pricing strategy based on data quality. Comput. Ind. Eng. 2017;112:1–10.
Zhang J, Zhang Z, Xiao X, Yang Y, Winslett M. Functional mechanism: regression analysis under differential privacy. Proc. VLDB Endow. 2012;5(11):1364–75.
Zhang N, Li M, Lou W. Distributed data mining with differential privacy. In: 2011 IEEE International Conference on Communications (ICC). IEEE; 2011. p. 1–5.
Zheng Z, Peng Y, Wu F, Tang S, Chen G. Arete: on designing joint online pricing and reward sharing mechanisms for mobile data markets. IEEE Trans. Mob. Comput. 2019;19(4):769–87.
Zhu R, Li Z, Wu F, Shin K, Chen G. Differentially private spectrum auction with approximate revenue maximization. In: Proceedings of the 15th ACM International Symposium on Mobile Ad Hoc Networking and Computing. ACM; 2014. p. 185–94.
Zhu T, Li G, Zhou W, Yu PS. Differentially private data publishing and analysis: a survey. IEEE Trans. Knowl. Data Eng. 2017;29(8):1619–38.



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